RFID Tool Tracking – Core Module Design

OBJECTIVE (WHAT THEY MUST DO)

Design a **working backend + basic UI flow** for:

Tool Issue / Return + Inventory Tracking System

📦SCOPE (STRICTLY LIMITED)

Only need to build:

✅**1. Tool Master**

✅**2. Issue / Return**

✅**3. Inventory Scan (simulate RFID)**

✅**4. Basic Dashboard**

🔹 1. TOOL MASTER (CRUD)

Each tool must have:

- Tool ID
- Name
- Category
- Status:
    - Available
    - Issued
    - Missing

🔹 2. ISSUE / RETURN FLOW

Issue Tool:

Input:

- Tool ID
- User ID

Logic:

- Check if available
- Mark as “Issued”

- Save transaction

Return Tool:

Input:

- Tool ID

Logic:

- Mark as “Available”
- Save transaction

🔹 3. INVENTORY SCAN (CORE TEST)

Simulate RFID scanning

Input:

- List of scanned Tool IDs

System should:

- Compare with database
- Identify:
    - Missing tools
    - Extra tools
    - Correct tools

🔹 4. DASHBOARD (BASIC)

Show:

- Total tools
- Issued tools
- Missing tools

TECH REQUIREMENTS

Let them choose stack (important for evaluation):

- MERN/ MEAN Stack
- UI : React + TypeScript
- Backend : Node.js + Express + TypeScript

- Database : MySQL/ PostgreSQL /Mongo DB

REQUIRED APIs with JWT Token

MUST design:

1. Create Tool

POST /tools

2. Issue Tool

POST /issue

3. Return Tool

POST /return

4. Inventory Scan

POST /scan

5. Dashboard

GET /stats

TEST CASES (MANDATORY)

You will check:

- Issue same tool twice → should fail
- Return non-issued tool → should handle
- Scan with missing tools → detect correctly
- Scan with extra tools → detect correctly

1. Working Demo

2. Database Schema

(Simple table structure)

3. Short Explanation (1–2 pages)

- How system works
- Assumptions
- Limitations